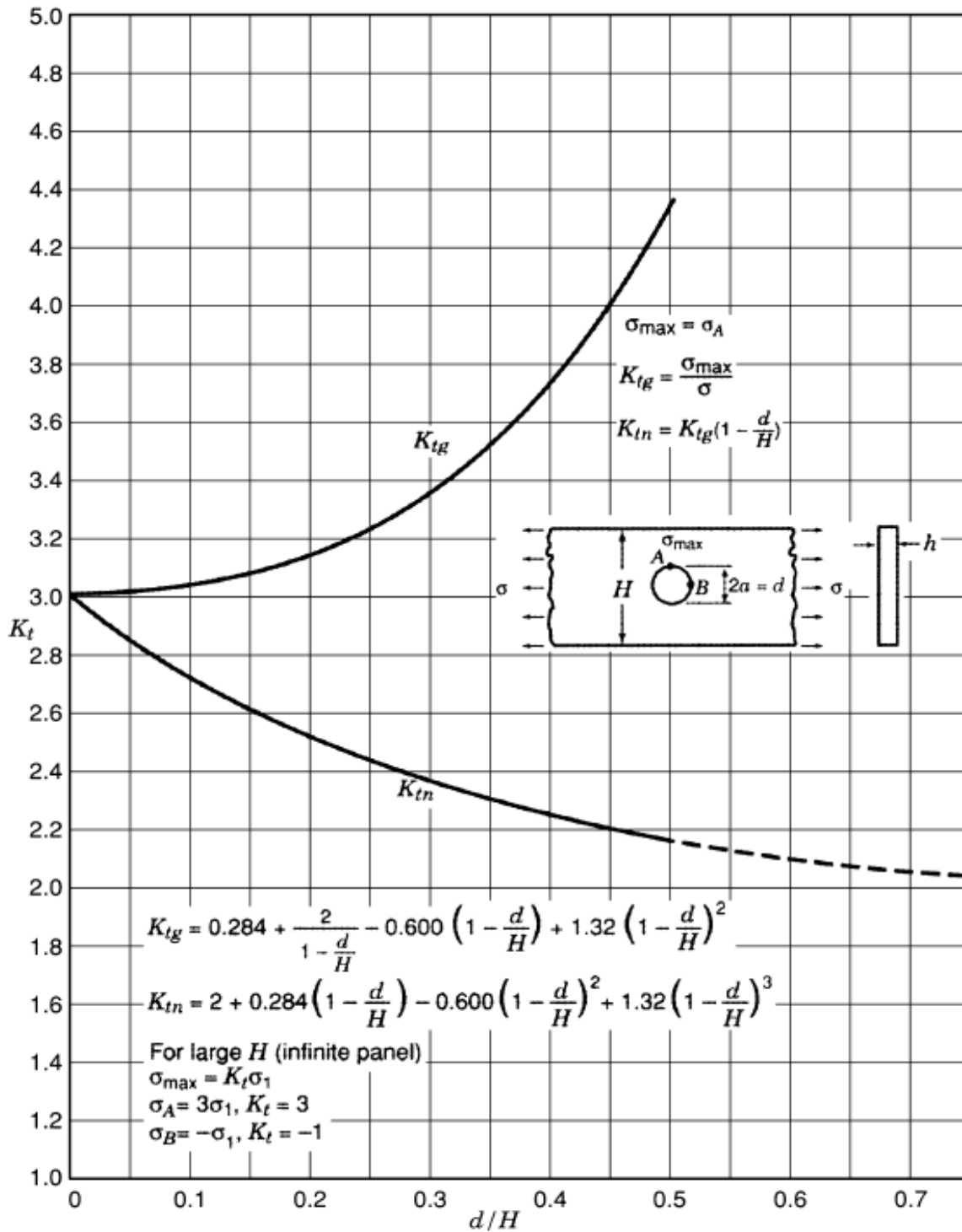
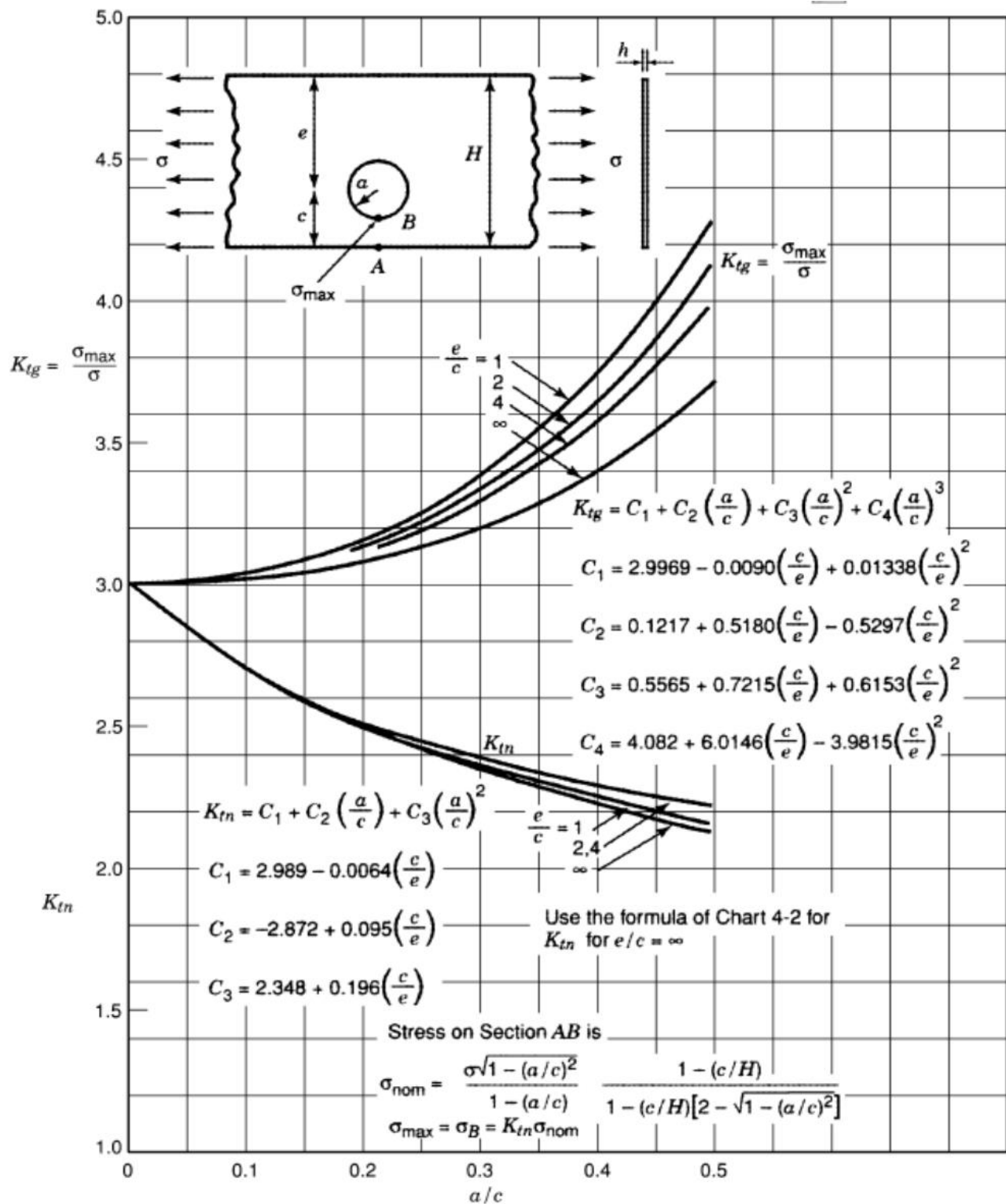


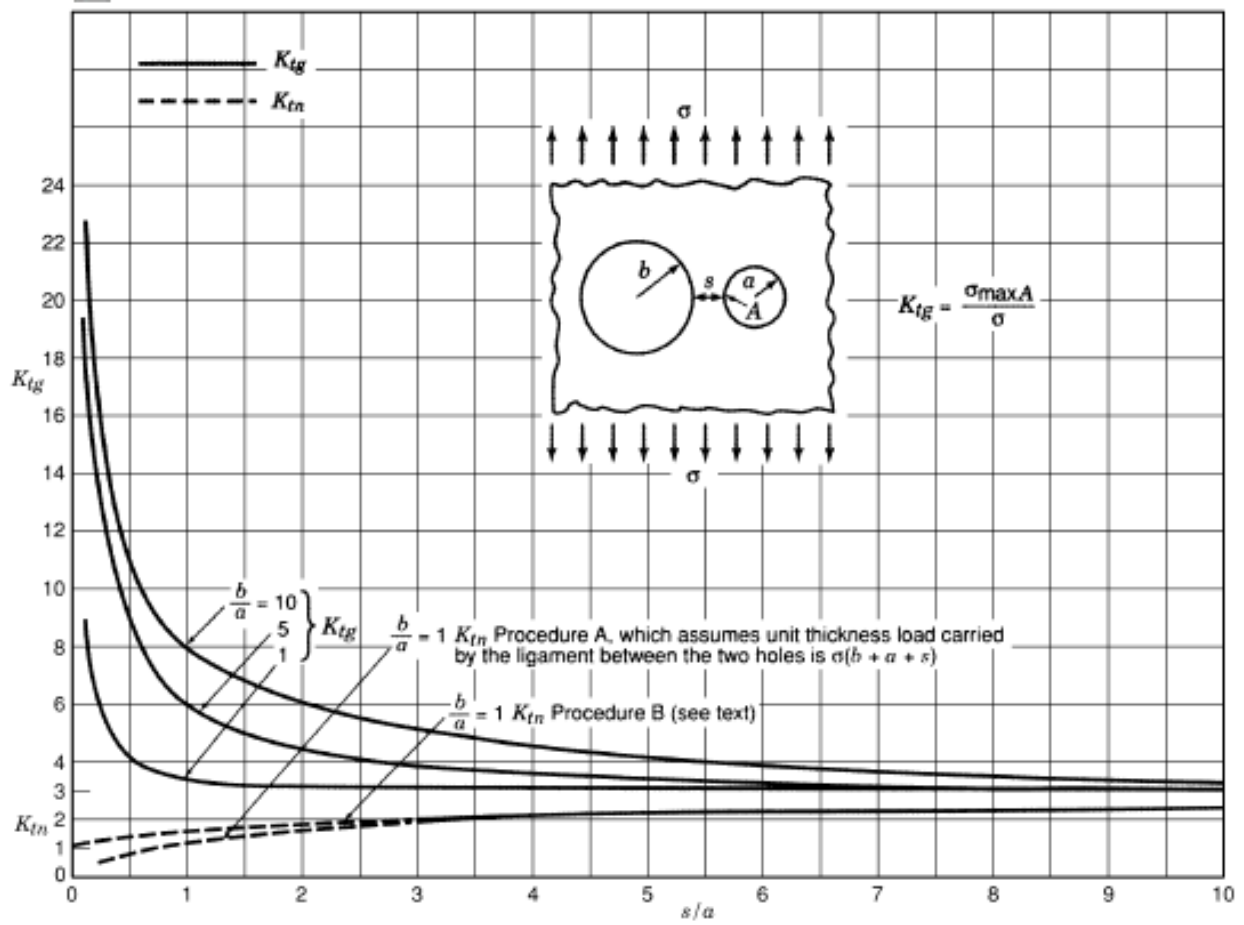
## PETERSON'S CHARTS



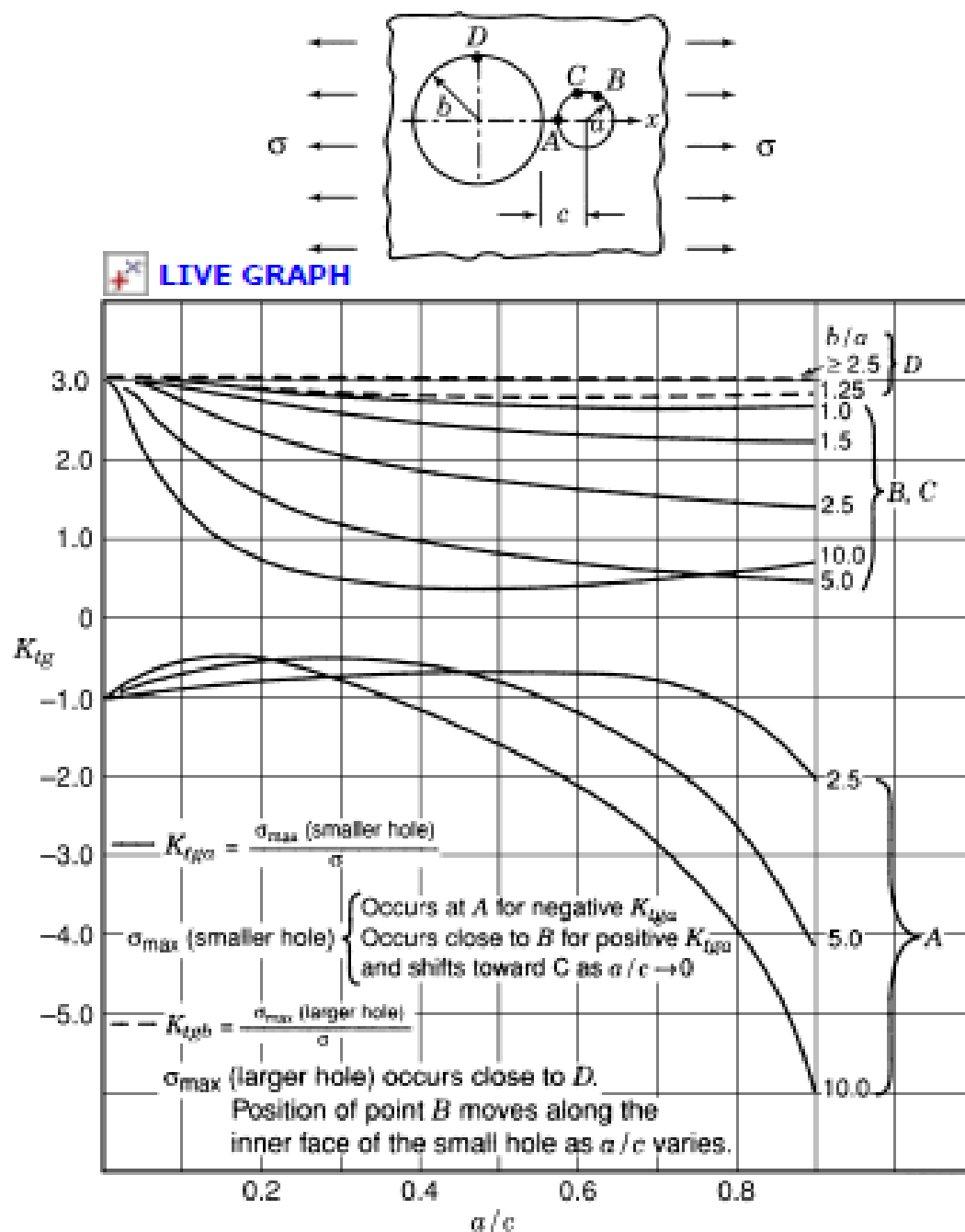
**Chart 4.1** Stress concentration factors  $K_{tg}$  and  $K_{tn}$  for the tension of a finite-width thin element with a circular hole (Howland 1929–30).



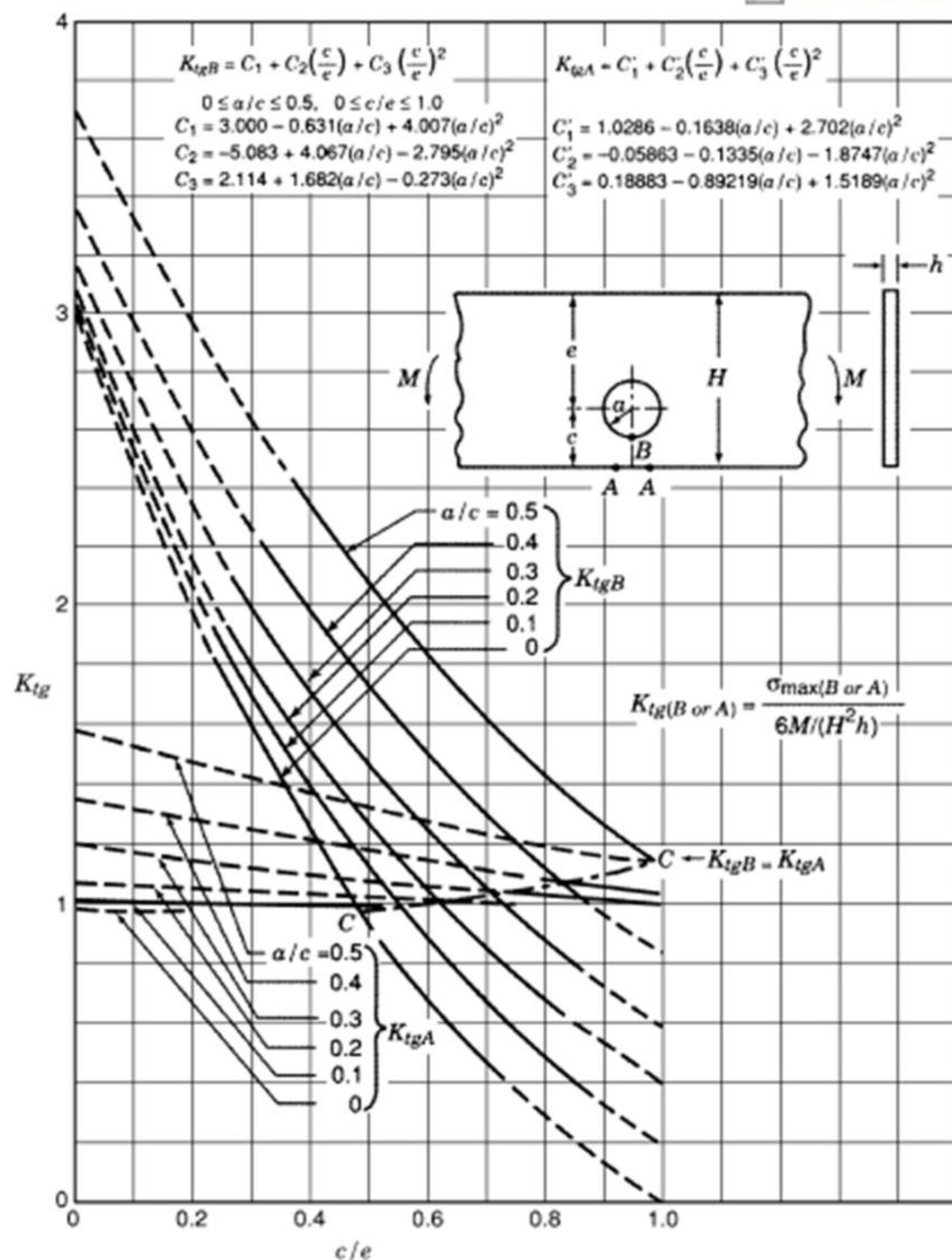
**Chart 4.3** Stress concentration factors for the tension of a finite-width element having an eccentrically located circular hole (based on mathematical analysis of Sjöström 1950).  $e/c = \infty$  corresponds to Chart 4.2.



**Chart 4.26** Stress concentration factors  $K_{tg}$  and  $K_{tn}$  for tension in an infinite thin element with two circular holes of unequal diameter (from mathematical analysis of Haddon 1967). Tension perpendicular to the line of holes.



**Chart 4.30** Stress concentration factors  $K_{tg}$  for tension in infinite thin element with two circular holes of unequal diameter (from mathematical analysis of Haddon 1967; ESDU 1981). Tension parallel to line of holes.



**Chart 4.80** Stress concentration factors  $K_{tg}$  for bending of a thin beam with a circular hole displaced from the center line (Isida 1952).

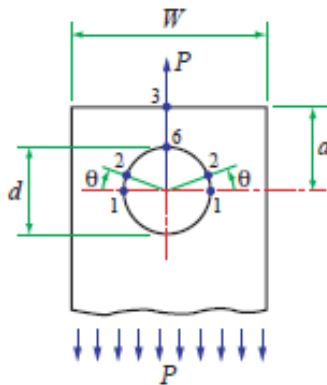
## STRESS CONCENTRATION FACTORS AXIALLY LOADED LUGS WITH CLEARANCE-FIT PINS

### 1. NOTATION

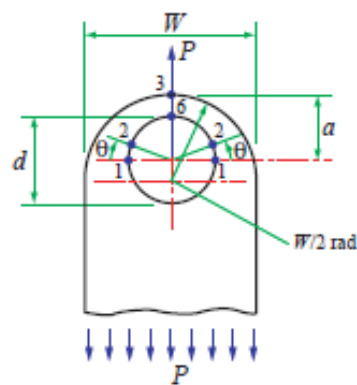
Both SI and British units are quoted but any coherent system of units may be used.

|                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                              |                  |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|
| $a$                                                                                                                                                                                                                                                                                                                                                                 | distance from centre of hole to lug end (see Sketches 1.1a and 1.1b)                                                         | m                | in                  |
| $d$                                                                                                                                                                                                                                                                                                                                                                 | diameter of hole                                                                                                             | m                | in                  |
| $e$                                                                                                                                                                                                                                                                                                                                                                 | pin-to-hole clearance as a percentage of $d$                                                                                 |                  |                     |
| $E_l, E_p$                                                                                                                                                                                                                                                                                                                                                          | moduli of elasticity of lug and pin materials, respectively                                                                  | N/m <sup>2</sup> | lbf/in <sup>2</sup> |
| $f_{max}$                                                                                                                                                                                                                                                                                                                                                           | maximum tensile stress in lug around hole perimeter                                                                          | N/m <sup>2</sup> | lbf/in <sup>2</sup> |
| $K'_e$                                                                                                                                                                                                                                                                                                                                                              | stress concentration factor at $e$ per cent clearance for lugs with $t/d$ less than 0.5, defined by                          | N/m <sup>2</sup> | lbf/in <sup>2</sup> |
| $f_{max} = K'_e f_{ref}$ <p>where <math>f_{ref} = \frac{P}{(W - d)t}</math></p> <p>(Thus <math>K'_{0.5}</math> relates to 0.5 per cent clearance between pin and hole and <math>K'_{100}</math> is used for the limiting case of point loading. Note that for round-ended lugs <math>f_{ref}</math> remains as defined here even when <math>a &lt; W/2</math>.)</p> |                                                                                                                              |                  |                     |
| $K''_e$                                                                                                                                                                                                                                                                                                                                                             | value of $K'_e$ when $t/d$ is greater than 0.5 (see Section 2.5)                                                             |                  |                     |
| $K_s$                                                                                                                                                                                                                                                                                                                                                               | stress concentration factor for waisted lug with flat and slot (related to $K_{wf}$ , see Section 2.4)                       |                  |                     |
| $K_w$                                                                                                                                                                                                                                                                                                                                                               | stress concentration factor for waisted lug (related to particular value of $K'_e$ , see Section 2.4)                        |                  |                     |
| $K_{wf}$                                                                                                                                                                                                                                                                                                                                                            | stress concentration factor for waisted lug with additional flat (related to a particular value of $K'_e$ , see Section 2.4) |                  |                     |
| $L$                                                                                                                                                                                                                                                                                                                                                                 | distance between lug hole and slot in waisted lug (see Section 2.4)                                                          | m                | in                  |
| $P$                                                                                                                                                                                                                                                                                                                                                                 | tensile load on lug applied via pin                                                                                          | N                | lbf                 |
| $t$                                                                                                                                                                                                                                                                                                                                                                 | thickness of lug (see Sketch 1.1c)                                                                                           | m                | in                  |
| $W$                                                                                                                                                                                                                                                                                                                                                                 | width of lug (see Sketches 1.1a and 1.1b)                                                                                    | m                | in                  |

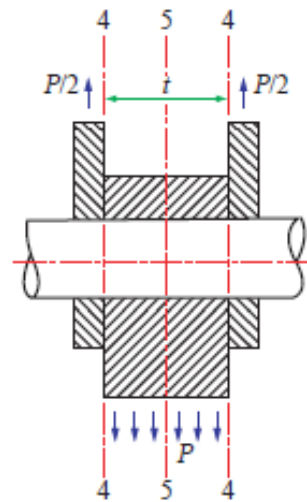
|          |                                                                        |        |        |
|----------|------------------------------------------------------------------------|--------|--------|
| $w$      | width of waisted lug (see Figures 5, 6 and 7)                          | m      | in     |
| $\eta_e$ | correction factor for clearance $e$ based on $K'_{0.2}$ and defined by |        |        |
|          | $\eta_e = \frac{K'_e - K'_{0.2}}{K'_{100} - K'_{0.2}}$                 |        |        |
|          | so that                                                                |        |        |
|          | $K'_e = K'_{0.2} + \eta_e(K'_{100} - K'_{0.2})$                        |        |        |
| $\theta$ | angle indicating position of $f_{max}$ (see Sketches 1.1a and 1.1b)    | degree | degree |



Sketch 1.1a Square-ended lugs



Sketch 1.1b Round-ended lugs



Sketch 1.1c Section through lug assembly centre line

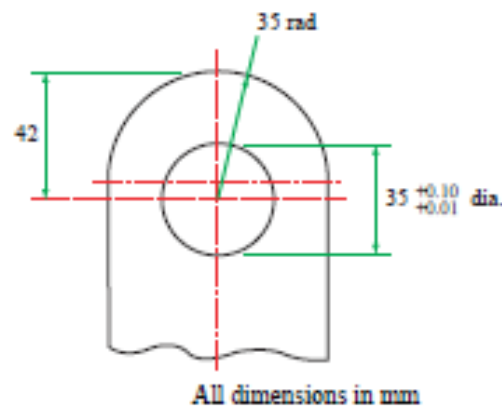
## 2. GENERAL NOTES

This Item provides the maximum elastic stress concentration factors at the perimeter of the hole in lugs loaded axially by single central clearance-fit pins. In Figures 1 to 3 data are provided for thin square- and round-ended plain lugs to which the following notes up to Section 2.3 apply. In Figures 5, 6, 7 and 8 data are given on particular examples of thin waisted round-ended lugs as guidance on the stress concentrations to be expected for such geometries. The notes in Section 2.4 qualify the use of the waisted lug data and their relationship to the data for plain lugs. In Figure 4 data on the effect of lug thickness are given, notes on the use of this figure being given in Section 2.5.

#### 4. EXAMPLES

##### 4.1 Example 1

It is required to estimate the maximum stress concentration factor acting at the lug shown in Sketch 4.1. The pin is to be  $35_{-0.01}^{+0.08}$  mm diameter.



Sketch 4.1 Lug of Example 1

The maximum clearance will be  $0.10 + 0.08 = 0.18$  mm.

Thus,

$$e = \frac{0.18 \times 100}{35} = 0.51.$$

Also,

$$\frac{d}{W} = \frac{35}{35 \times 2} = 0.5$$

and 
$$\frac{a}{W} = \frac{42}{35 \times 2} = 0.6.$$

Thus, from Figure 2 for round-ended lugs,  $K'_{0.2} = 2.68$  and  $K'_{100} = 3.61$ .

From Figure 3,  $\eta_e = 0.22$  so that, from the definition of  $\eta_e$  given in the notation, the maximum stress concentration is estimated as

$$K'_e = 2.68 + 0.22(3.61 - 2.68) = 2.88.$$

Repeating the calculation for the minimum clearance ( $e = 0.06$ ) gives a minimum stress concentration of  $K'_e = 2.46$ .



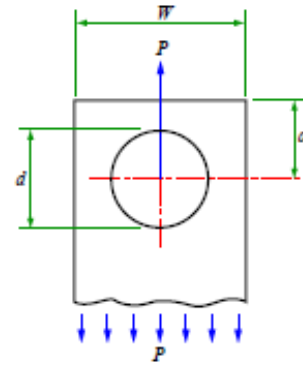
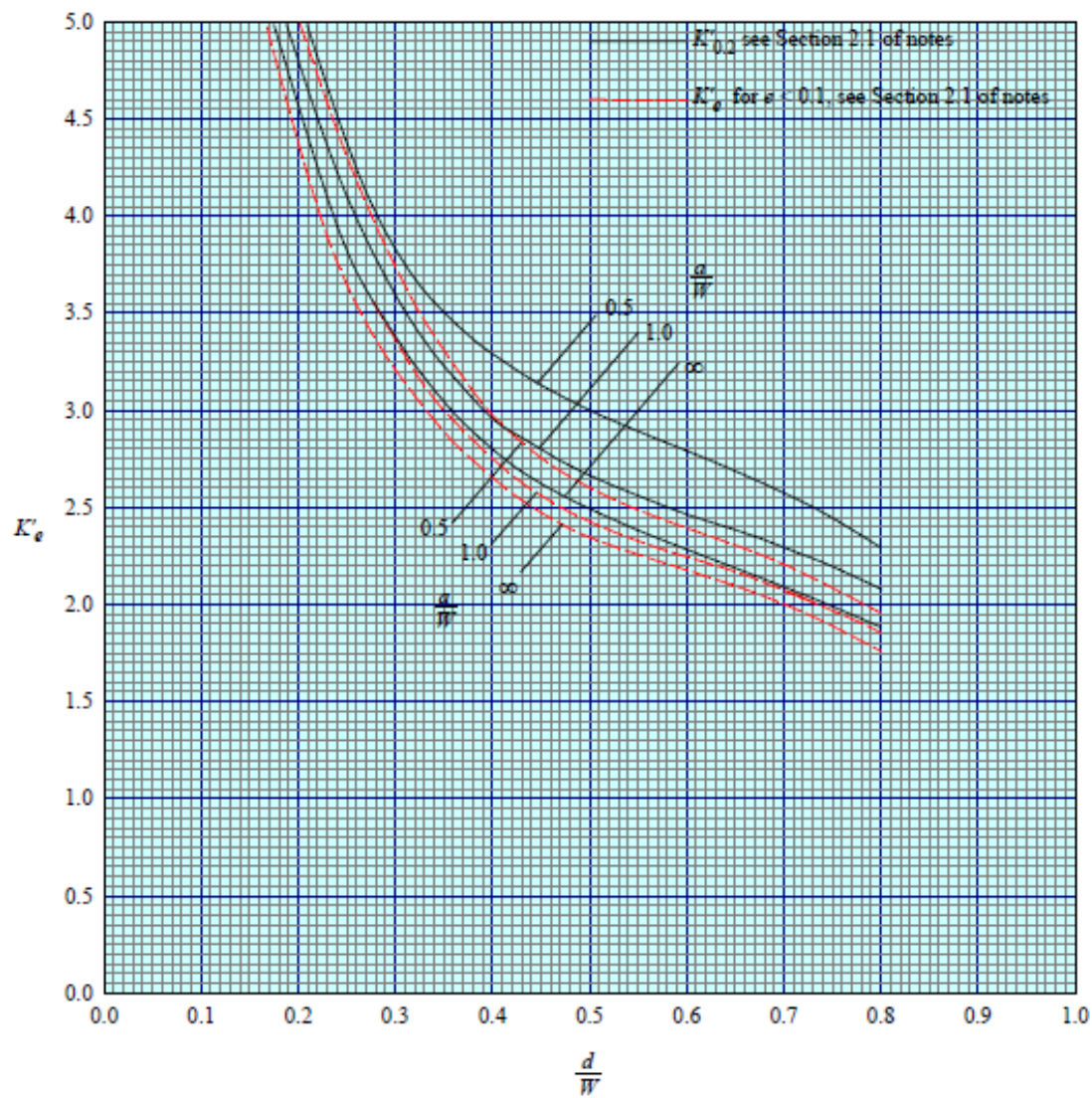


FIGURE 1 SQUARE-ENDED LUGS

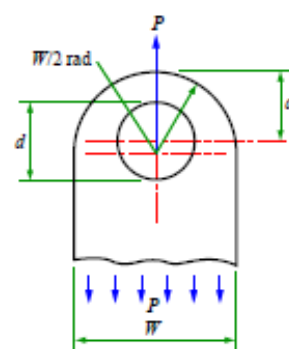
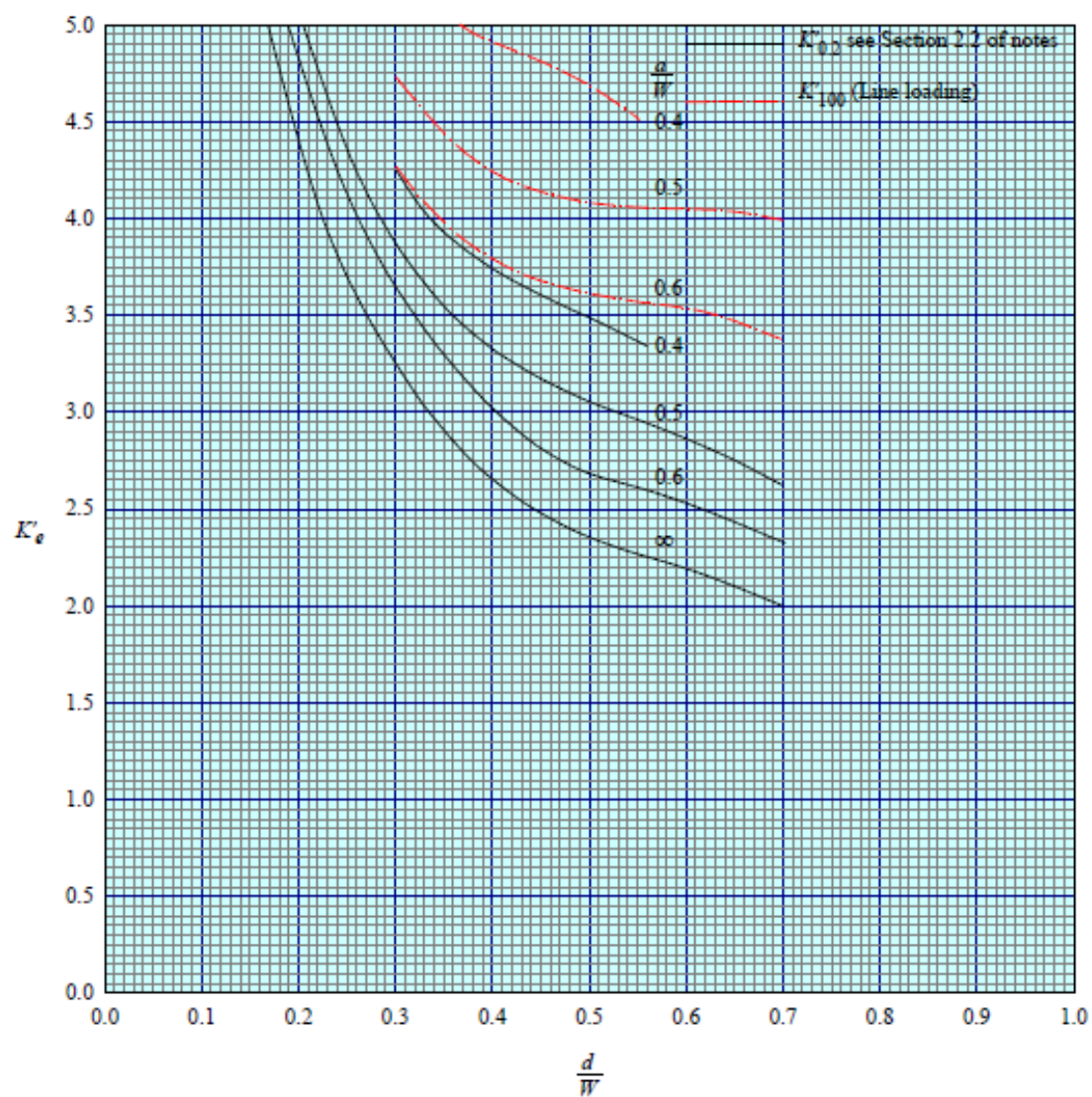


FIGURE 2 ROUND-ENDED LUGS

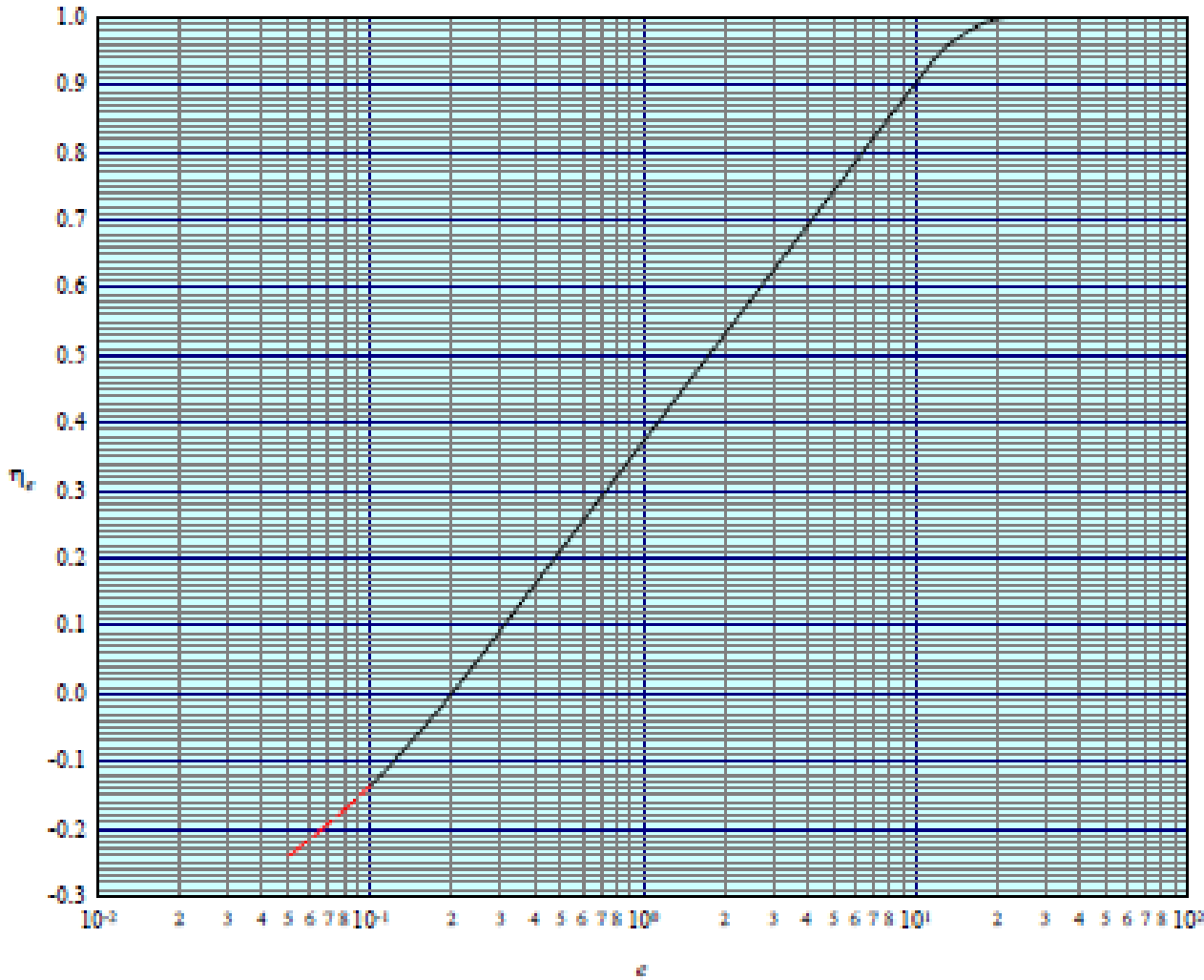


FIGURE 3 CORRECTION FACTOR FOR CLEARANCE

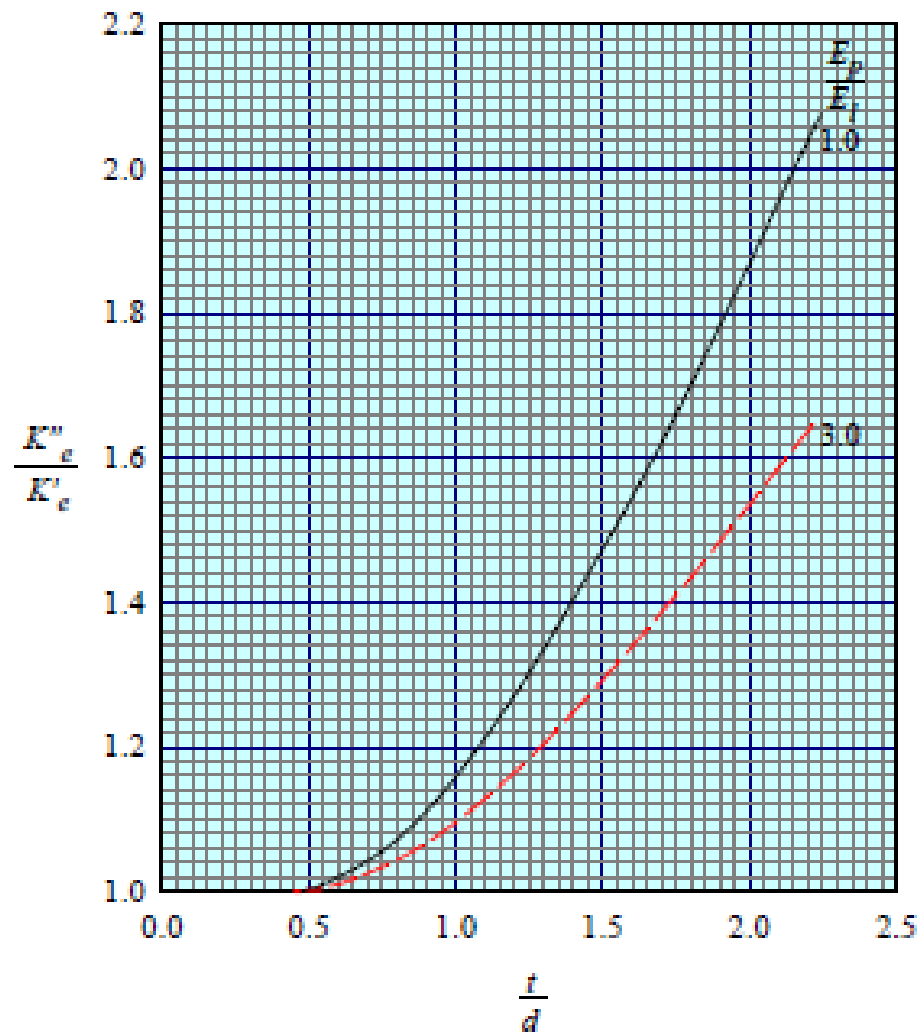


FIGURE 4 EFFECT OF LUG THICKNESS